

Sambasiva Rao Pendela
Sr. Frontend Developer & Agentic AI Architect

PROFESSIONAL SUMMARY:

AI & Frontend Architect with 15+ years of experience in designing and delivering intelligent Web and Mobile Applications. I specialize in Artificial Intelligence, Generative AI, Agentic AI, and modern JavaScript frameworks, enabling scalable and innovative digital solutions. I possess strong expertise in solution architecture, system design, cloud technologies, and Software Development Life Cycle (SDLC) methodologies, including development, testing, deployment, and maintenance. Throughout my career, I have successfully led cross-functional teams, architected enterprise-grade platforms, and delivered high-quality solutions from concept to production. I thrive in driving innovation through AI-powered technologies while staying at the forefront of emerging industry trends and best practices.

- Possess strong knowledge of **Machine Learning** concepts including Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, and Data Preprocessing techniques for building intelligent applications.
- Experienced in implementing **Deep Learning** solutions using Neural Networks, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, and Attention Mechanisms for solving complex business problems and predictive analytics.
- Proficient in **Generative AI** technologies, including Large Language Models (LLMs), Prompt Engineering, Embedding Models, Vector Databases, Retrieval-Augmented Generation (RAG), Fine-Tuning, and AI-powered application development.
- Hands-on experience in designing and developing **Agentic AI Systems** utilizing Multi-Agent Architectures, Autonomous Task Execution, Tool Calling, Memory Management, Planning and Reasoning Workflows, and Human-in-the-Loop AI solutions.
- Skilled in leveraging modern AI frameworks and platforms such as **LangChain, LangGraph, CrewAI, AutoGen, FastAPI, OpenAI APIs, MCP (Model Context Protocol), Docker, Kubernetes, and Cloud Services** to build scalable enterprise-grade AI applications.
- Proficient in JavaScript ES-6, demonstrating advanced knowledge of concepts such as arrow functions, object destructuring, block-scoped constructs, template literals, promises, multi-line strings, as well as ES-7 async functions, observables, and object rest/spread properties. This deep understanding allows for efficient and elegant JavaScript development.
- Possess hands-on experience in utilizing the JAX-WS API and Apache CXF framework for SOAP implementation, while also proficiently automating documentation using Swagger and WSDL.
- Proficient in working with various version control tools including Git, SVN, Bitbuckets, TFS, and CVS, as well as diverse Project management tools like JIRA, Rally, and Microsoft Project.
- Proficient in developing user interfaces utilizing a range of technologies such as HTML5, CSS3, and JavaScript. Developed applications with AJAX to handle asynchronous calls, adhering to the established W3C Web standards.

- Skilled in developing applications using Node.js Socket.io programming to establish persistent connections between clients and servers.
- Extensive hands-on experience with the Node.js platform, utilizing NPM for library management and BOWER for dependency management.
- Well-versed in utilizing JavaScript testing frameworks such as Karma, Jasmine, Protractor for Angular 16, and Mocha and Chai for React JS and Node.js.
- Exhibit a strong aptitude for documentation, coupled with a proven track record of working within an agile-based delivery model.
- Demonstrates adeptness in making informed and impactful decisions, coupled with a skillful ability to navigate through change. Exhibits exemplary communication skills in both written and verbal formats within a professional context.
- Exemplifies a collaborative mindset, showcasing a positive demeanor, unwavering enthusiasm, self-driven initiative, and inherent self-motivation.
- Proficiently adept at multitasking, radiating energy, and displaying a rapid grasp of new concepts, while also possessing a natural aptitude for creative problem-solving.
- Extensive experience with Docker containers, effectively leveraging their capabilities to streamline application deployment and foster collaboration and innovation within hybrid cloud environments.
- Well-versed in implementing Model-View-Controller (MVC) architecture on the client side, utilizing frameworks such as AngularJS, React with Redux, React with Flux, Backbone JS, Ext JS, and Knockout JS.
- Deep understanding and proficiency in implementing Model-View-View Model (MVVM) architecture on the client side, employing Angular 16 and newer versions in conjunction with TypeScript 4.X.
- Comprehensive understanding of Amazon Web Services (AWS), including Amazon EC2, Amazon S3, Amazon Simple DB, Amazon Elastic Load Balancing, Amazon SQS, Amazon EBS, and Amazon CloudFront.
- Expertise in creating instances using Amazon EC2 and proficient in serverless programming with AWS Lambda. Skilled in utilizing AWS Elastic Beanstalk for efficient capacity provisioning and monitoring application health.
- Experienced in designing and implementing enterprise-grade AI architectures that integrate Large Language Models (LLMs), Machine Learning models, and business applications to deliver intelligent automation solutions.
- Proven expertise in architecting scalable multi-agent ecosystems capable of autonomous decision-making, task orchestration, and seamless integration with enterprise systems and third-party services.
- Adept at leading cross-functional teams in the adoption of AI technologies, establishing AI best practices, defining architecture standards, and driving innovation through emerging AI capabilities.
- Strong understanding of AI governance, responsible AI practices, model evaluation techniques, security considerations, and ethical deployment of AI solutions in enterprise environments.
- Skilled in developing AI-powered conversational systems, virtual assistants, and knowledge management platforms utilizing NLP, Generative AI, and Agentic AI technologies.

TECHNICAL SKILLS

AI Stack:	Python, ML, DL, NLP, Generative AI and Agentic AI
Vector Databases:	Pinecone, Chroma DB, FAISS, Weaviate, Qdrant
RAG Technologies:	Retrieval-Augmented Generation (RAG), Knowledge Retrieval, Document Processing, Chunking Strategies, Semantic Search
Programming Languages:	Java, C, C++, PL/SQL, .Net
Web Technologies:	JavaScript, HTML5, CSS3, AJAX, JSON, Bootstrap, jQuery, Angular2.0/4.0, React JS with Flux, Redux, Ember JS, Knockout JS, Backbone JS, Handlebar JS, NodeJS, Express JS, MEAN& MERN Stack.
Databases:	Oracle 8i/9i/10g/11i, DB2, MongoDB, Cassandra, SQL Server 2008, MYSQL
Software Methodologies:	Agile, Waterfall, Scrum, TDD
Build & Configuration Tools:	Jenkins, Ant, Maven, Gradle, Bamboo Gulp, Grunt, Webpack, NPM, Bower.
Testing Tools:	Junit, Mockito, Jasmine, Karma, Mocha, Chai, Spock, Protractor
Version Control Tools:	GIT, Bitbucket, TFS, SVN, CVS
Project Management Tools:	JIRA, Rally, Microsoft Project
Cloud Technologies and DevOps Tools:	AWS, PCF, EC2, ECS, EBS, EKS, EBS, S3, Docker, SQS, RDS, S3, Lambda, Kubernetes.
Operating System:	Windows, Unix, Linux, MAC, MS-DOS
IDE's:	Visual Studio, Eclipse, IntelliJ, Spring Tool Suite

PROFESSIONAL EXPERIENCE

Client: Brane Enterprises

Aug 2023 – Present

Role: Solution Leader

Project: Brane Education

Description:

Developed a cross-platform educational app that integrates voice-based inputs, AI-driven personalization, and data visualization. The app features voice recognition for enhanced user interaction and accessibility, while AI algorithms deliver personalized learning experiences tailored to user performance. Interactive graphs were designed for real-time tracking of student progress, supported by a scalable content management system.

Responsibilities:

- Built a robust backend to manage educational content, enabling easy updates and scalability.
- Utilized the SCRUM framework to facilitate the development of the new system, adhering to agile software development principles.
- Implemented seamless voice-based inputs, enhancing user interaction and accessibility for diverse learners.
- Integrated AI algorithms to deliver personalized learning experiences, adapting content based on user performance and preferences.
- Designed and implemented interactive graphs to track student progress and visualize learning outcomes in real-time.
- Incorporated gamification elements, such as badges and leaderboards, to motivate and engage users.
- Integrated a feedback mechanism for users to provide real-time input on their learning experience, enabling continuous app enhancement.
- Established a reliable and efficient CI/CD pipeline utilizing Docker, Jenkins, and Azure Container Service. This pipeline facilitated the seamless deployment of applications, ensuring consistent delivery of new features and updates.

Environment:

Python, Machine Learning, Deep Learning, Generative AI, Agentic AI, Large Language Models (LLMs), LangChain, LangGraph, CrewAI, AutoGen, OpenAI GPT, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone, ChromaDB, FAISS), Prompt Engineering, AI Agents, Multi-Agent Systems, FastAPI, REST APIs, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Jupyter Notebook, Apache Airflow, Docker, Jenkins, Kubernetes, Azure AI Services, AWS Bedrock, Azure OpenAI, Git, GitHub, JIRA, Agile/SCRUM.

Client: Sathya Technologies

Sep 2020 – July 2023

Role: MERN Stack Architect and Training Head

Project: Ari Mobile Calibrations

Description:

The React Native Application Designed to read the Data from Bluetooth (HC-05) in a Specified Intervals and Reads the GPS Co-ordinates and Generate the Report on Google Map with data collected from Bluetooth Device and Google APIs.

Responsibilities:

- Utilized the SCRUM framework to facilitate the development of the new system, adhering to agile software development principles.
- Employed React & React Native for the rendering of the CTRS Web pages, leveraging its powerful capabilities.
- Crafted and executed a Single Page Application (SPA) using React Native and React18.X Developed reusable TypeScript components and services to seamlessly consume REST APIs, leveraging the component-based architecture provided by Angular 6. These services were designed with async capabilities and included comprehensive descriptions of observables.
- Implemented state management using Redux patterns, allowing for effective management of data UI through the utilization of stores, while ensuring smooth communication with backend services using effects and reducers.
- Within the SCRUM framework, conducted daily status meetings to keep the team aligned and informed. Additionally, adopted a Test-Driven Development (TDD) approach using the JUnit Mockito framework.
- Implemented security APIs incorporating JWT access tokens and API keys to facilitate secure access for external consumers.
- Developed custom elements, HTML imports, and Shadow DOM using Angular 6, ensuring complete encapsulation, and implemented Angular Router to enable smooth navigation between views as users perform application tasks.
- Implemented authentication and authorization mechanisms using Spring Security and OAuth2.0 to ensure secure access to the application.
- Established a reliable and efficient CI/CD pipeline utilizing Docker, Jenkins, and Azure Container Service. This pipeline facilitated the seamless deployment of applications, ensuring consistent delivery of new features and updates.

Environment:

HTML5, CSS3, Ajax, Bootstrap 5.X, React, React Native JSON, OAuth 2.0, JUnit, Mockito, Maven, Kafka, Cassandra, RESTful, SOAP, JIRA

Client: Naresh I Technologies

Oct 2015 – Aug 2020

Role: Sr. Web Technologies Architect & Trainer.

Project: OMS – Homepage Approval Notification Widget

Description:

The widget app was designed to provide alerts to employees based on the requests they create. The app will listen for changes to an xml file containing employee details on a constant basis using the node module Socket.IO. Any changes to the xml file will trigger a 'emit' event and the Change will subsequently be displayed to the UI using ReactJS.

Responsibilities:

- Proficient with container systems like Docker and container orchestration like EC2 Container Service, Kubernetes, worked with Terraform.
- Managed Docker orchestration and Docker containerization using Kubernetes.
- Used Kubernetes to orchestrate the deployment, scaling and management of Docker Containers.
- Redux/NGRX and understanding of the Redux pattern
- Wrote Client-Side code using React JS and used NPM, directories to generate the server environments used flux to manage with targeted URL's (Unidirectional data flows) and to monitor the application.
- REACT JS Virtual DOM used for client-side view rendering services, React-Redux for state management and React-Router for programmatic navigation.
- Involved in the creation of UI specification documents by interacting with business users and design documentation and data modelling for the entire project.
- Expertise in using React.js/Redux to build User Interface, strong knowledge of state store, middleware, action creator, reducer and container.
- Written several Axios service calls like get () and post () to make REST service calls.
- Developed user interface by using ReactJS, Redux for SPA development and implemented client-side Interface using ReactJS.
- Proficient in writing react code with help of ECMAScript (ES6) and JSX. Implemented Virtual Dom for Client-Side view rendering services using ReactJS.
- implemented platform modules through spring Java micro and Node.js services by employing Hapi.js, as well as front-end React JS.
- Worked with Web pack for bundling application source code in convenient chunks and for loading that code from a server into a browser
- Implemented unit test cases using Karma and Jasmine on client side.
- Created reusable templates using Angular directives and worked with NPM package manager tools (Node JS).

Environment:

AWS, AWS Load Balancer, EC2, Amazon Dynamo DB, Swagger, NodeJS, ReactJs, Web pack, ECMA6, Karma and jasmine, Spring 4.x, Spring Boot, ZUUL, Micro services architecture, Docker, Cassandra DB, OAUTH

Client: Santosh Technologies

Jan 2014 – Nov 2014

Role: Android Developer & Trainer

Project: SMART IO

Description:

A mobile application is developed using android to generate Pulse Width Modulated signal with desired frequency and duty cycle which can be varied, digital switch implementation and getting the digital data using ATMEGA 328p controller. The main aim of the project is to receive the data from the mobile application using Bluetooth module HC-05 and generate digital output or Pulse Width Modulated Signal or return a digital value of an analog signal to application.

Responsibilities:

- Understand the app requirements and design it from scratch.
- Follow MVC design pattern and ensure correct coding standards.
- Involved in testing the app and resolving issues
- Work on app improvements and feedback from concerned stakeholders.

Environment:

Android

Client: Durga Software Solutions

September 2008 – Dec 2013

Role: Android Developer & Trainer

Project: Durga Education

Description:

Durga Education App contains mainly four features such as Engineering colleges info, Notifications, Results and Jobs. Engineering college's info feature provides the information related to the engineering colleges, Notifications feature displays the notifications released by the universities, Results feature provides the students results released by the universities and Jobs feature helps to search the jobs easily all over India.

Responsibilities:

- Understand the app requirements and design it from scratch.
- Follow MVC design pattern and ensure correct coding standards.
- Involved in testing the app and resolving issues
- Work on app improvements and feedback from concerned stakeholders.
- Supporting The Team Members for Resolving complicated issues.
- Responsible for validating controls and fixing bugs

Environment:

Android

CORPORATE TRAININGS

- Conduct on-site training as per client requirements.
- Coordinate and conduct small group and individual training sessions based on staff and organizational learning needs.
- Work with corporate leadership to assess existing courses and develop programs to fill gaps.

Client Name	Technology	Organized By
Indian Institute of Technology Bombay	Full Stack	ExcelR
Align Technology	Agentic AI & Generative AI	Align Technology Hyderabad
Benfranklin	MERN Stack with AI	Benfranklin Hyderabad
Sankalpa Software Solutions	MEAN Stack with Gen AI	Hyderabad
Gait View Technophiles Pvt. Ltd	Android	Vijayawada

DELIVERED EXPERT LECTURES

College Name	Technology	Students
Bannari Amman Institute of Technology, Tamil Nadu	WEB TECHNOLOGIES	CSE& IT Staff
Sphoorthy Engineering College Hyderabad	ANGULAR	Staff
Sphoorthy Engineering College Hyderabad	REACTJS	Staff

EDUCATION

2008: B. Tech JNTU Kakinada 1st Grade